

IMPERIAL

Mesh coarsening

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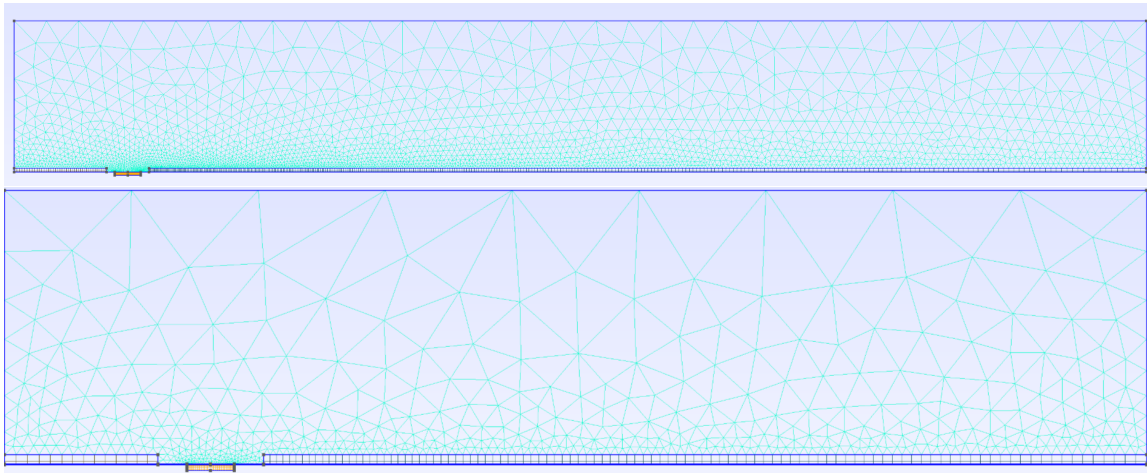
Old Mesh vs New mesh ($d = 3$, $w = 26$)

- Old mesh: $p = 5$, 11673 elements.
- New mesh: $p = 5$, 2328 elements.

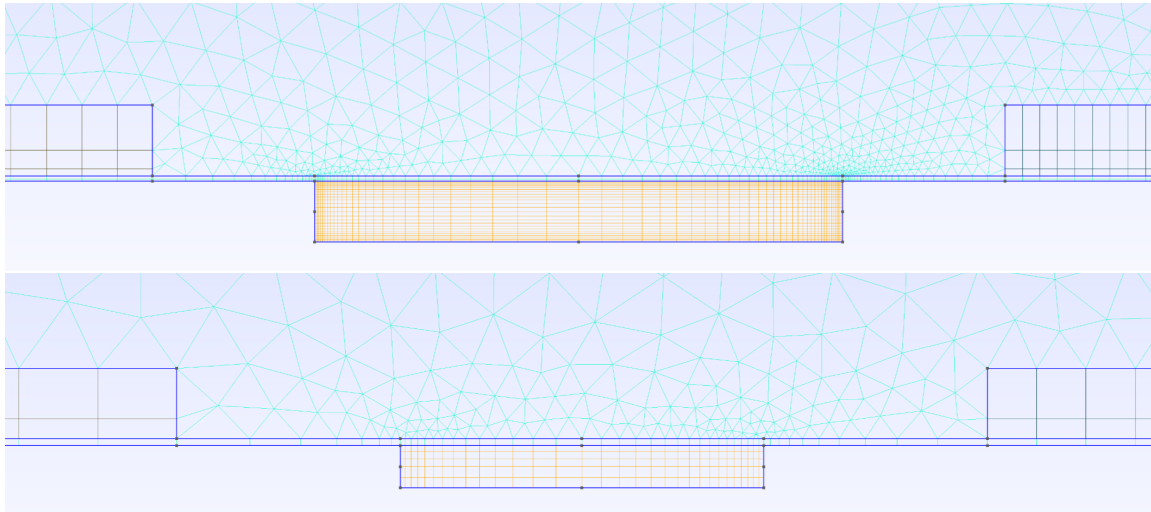
Most important implementation is that I shortened the downstream region from $1000\delta^*$ to $500\delta^*$. Initially Jeff told us to use a domain such that at the outflow we have $Re_{\delta^*} = 2000$, but even for the n-factor computation I think with half the length we can still have a good approximation of the n-factor curves.

I mean, for the compressible simulations I was already using $500\delta^*$.

Mesh comparison (zoomed out)



Mesh comparison (zoomed in)



Physics comparison

- L 2 error (variable u): 304.644 (old) vs 304.646 (new)
- L inf error (variable u): 1.01793 (old) vs 1.02519 (new)
- L 2 error (variable v): 0.510122 (old) vs 0.590828 (new)
- L inf error (variable v): 0.115768 (old) vs 0.157218 (new)
- L 2 error (variable p): 0.1737 (old) vs 1.3124 (new)
- L inf error (variable p): 0.0527884 (old) vs 0.0949722 (new)
- ω : 0.263935 (old) vs 0.264117 (new)

Pressure comparison

